Data Collection Report

1. Book Dataset Collection

Link to the collected data: Book Mood Annotation Questionnaire

For my Book Mood Reader project, I created a questionnaire to collect my own dataset of books and their associated moods. The goal of this dataset is to train a model that can recognize the mood of a book based on information such as its title, author and description.

The questionnaire asks participants to provide books they have read and to assign mood labels that they believe describe the atmosphere of each book. For the project, I created a predefined set of **25 mood labels**, which are used consistently across the collected responses so that the data can later be transformed into a structured dataset suitable for machine learning.

The collected information includes the book title, author, a short description and the selected mood labels with their assigned scores. Collecting the data through readers is useful because the mood of a book can be subjective, and people who have actually read the book can provide a more realistic interpretation of its atmosphere.

The book descriptions will later be transformed into numerical features using **TF-IDF**, which allows the model to work with the text and learn patterns related to the 25 mood labels.

During the collection process, I noticed a small mistake in the way I structured the mood-ranking part of the questionnaire. Because of this, some responses contain more mood information than I actually need for the first version of the model. During the data-cleaning stage, I will compare the assigned scores and keep only the mood label or labels with the highest relevant score for each book. This will make the final dataset more consistent and easier to use for training and evaluating the model.

Before training, I will also check the dataset for duplicated books, missing information, inconsistent spelling and other values that may need to be standardized.

2. Song Dataset Collection

The second dataset contains a collection of downloaded songs that I will use for the music side of the Book Mood Reader system.

For this dataset, I used the songs I had already downloaded and manually assigned mood labels to each of them. I used **the same set of 25 mood labels as in the book dataset**. Keeping the exact same labels for both datasets is important because it allows the book and music data to be compared directly later in the system.

For example, if a book is labelled as *romantic*, *dreamy* and *melancholic*, songs in the second dataset can also contain these same labels. This makes it possible to compare the predicted moods of a book with the moods assigned to songs and find the closest matches.

For some of the song selection, I also used songs from publicly available playlists connected to books. These include playlists suggested by authors or playlists related to specific books. This gives the music dataset a stronger connection to the book domain, since some of

the selected songs were already associated with the atmosphere or inspiration behind particular books.

The final song dataset therefore contains the downloaded audio files together with manually assigned mood labels from the same set of 25 labels used for the books. Later, the audio files can also be analysed using **Librosa** to extract audio features that can support the song mood classification.

By using the same 25 mood labels in both datasets, the final system can compare book moods and song moods, calculate their similarity, rank the songs and generate a playlist that matches the atmosphere of a selected book.